

Discrete Mathematics

Exercises — Coding Theory (Set 5b)

Conventions. All codes are binary (over $\mathbb{F}_2$). For a code with minimum distance $d_{\min}$, it *detects* up to $d_{\min} - 1$ errors and *corrects* up to $t = \lfloor (d_{\min} - 1)/2 \rfloor$ errors. A linear $[n, k, d_{\min}]$ code is *perfect* if $2^k \cdot V(n, t) = 2^n$, where $V(n, t) = \sum_{i=0}^t \binom{n}{i}$. The *syndrome* of a received word $\mathbf{r}$ with respect to parity-check matrix H is $\mathbf{s} = H\mathbf{r}^\top$; a nonzero syndrome identifies the error position (for single-error correction) as the column of H equal to $\mathbf{s}$.

Exercise 1

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

- (a) List all codewords of this code.
- (b) Find a parity-check matrix H .
- (c) Determine $d_{\min}$.
- (d) How many errors does this code detect? How many does it correct?
- (e) Is this code perfect?

Exercise 2

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .
- (b) List all codewords of this code.
- (c) Determine $d_{\min}$.
- (d) How many errors does this code detect? How many does it correct?
- (e) Is this code perfect?

Exercise 3

A binary linear code has parity-check matrix

$$H = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find a generator matrix G for this code.
- (b) List all codewords.
- (c) Determine $d_{\min}$.
- (d) How many errors does this code detect? How many does it correct?
- (e) Is this code perfect?

Exercise 4

The *5-fold repetition code* encodes a single information bit $u \in \mathbb{F}_2$ as the codeword $uuuuu$.

- (a) Write down a generator matrix G for this code and list all codewords.
- (b) Determine $d_{\min}$.
- (c) How many errors does this code detect? How many does it correct?
- (d) Is this code perfect? (*Hint: compute $V(5, t)$ for the appropriate t .*)

Exercise 5

Let $\mathcal{C} = \{\mathbf{0}, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$ where

$$\mathbf{c}_1 = (1, 0, 1, 1, 0, 1), \quad \mathbf{c}_2 = (0, 1, 0, 1, 1, 1), \quad \mathbf{c}_3 = (1, 1, 1, 0, 1, 0).$$

- (a) Verify that $\mathcal{C}$ is a linear code.
- (b) Write down a generator matrix G .
- (c) Find a parity-check matrix H .
- (d) Determine $d_{\min}$.
- (e) How many errors does this code detect? How many does it correct?
- (f) Is this code perfect?

Exercise 6

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .
- (b) Determine $d_{\min}$.
- (c) How many errors does this code detect? How many does it correct?
- (d) Is this code perfect?
- (e) The word $\mathbf{r} = (1, 0, 1, 0, 0, 1, 1)$ was received. Assuming at most one transmission error occurred, compute the syndrome $\mathbf{s} = H\mathbf{r}^\top$, identify the error position, and recover the sent codeword.

Exercise 7

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .
- (b) List all codewords of this code.
- (c) Determine $d_{\min}$.
- (d) How many errors does this code detect? How many does it correct?
- (e) Is this code perfect? Compare your answer with Exercise 6: what relationship do you observe between these two codes?

Exercise 8

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .

- (b) Determine $d_{\min}$. (*Hint: compute the weights of the four generator rows and a few of their sums.*)
- (c) How many errors does this code detect? How many does it correct?
- (d) Is this code perfect?
- (e) The word $\mathbf{r} = (1, 0, 0, 1, 0, 1, 0, 0)$ was received. Assuming at most one transmission error occurred, use syndrome decoding to recover the sent codeword.

Exercise 9

A binary linear code has parity-check matrix

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find a systematic generator matrix $G = [I_4 \mid P]$ for this code.
- (b) The word $\mathbf{r} = (0, 1, 0, 1, 1, 1, 0)$ was received. Assuming at most one transmission error occurred, compute the syndrome $\mathbf{s} = H\mathbf{r}^\top$, identify the error position, and recover the sent codeword.
- (c) Identify the message $\mathbf{u} \in \mathbb{F}_2^4$ that was originally encoded.

Exercise 10

Consider the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

- (a) Find a parity-check matrix H .
- (b) Determine $d_{\min}$, the number of detectable errors, and the number of correctable errors.
- (c) The word $\mathbf{r} = (1, 0, 0, 1, 1, 1)$ was received. Compute the syndrome and, assuming at most one error occurred, correct $\mathbf{r}$ to the nearest codeword $\hat{\mathbf{c}}$.
- (d) Recover the message $\mathbf{u}$ that produced $\hat{\mathbf{c}}$.
- (e) Explain why a received word with zero syndrome might still contain errors. Under what minimum condition on $d_{\min}$ would a two-error pattern go undetected?